

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Extended Reality Technologies [NPT_4135]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What is the key difference between kinematics and dynamics in physical modeling? (2)
 - A)
 - B) Write one major challenge in creating a Multiuser Virtual Environment (e.g., latency, consistency). (3)
 - C) Explain the concept of Level of Detail (LOD) as a model management technique. Why is it used, and how does it improve rendering performance? (5)
- 2) List out the primary function of a "Digital Twin" in an industrial XR context? (2)
 - A)
 - B) Why is optimization particularly crucial for VR/AR experiences compared to traditional desktop games? (3)
 - C) Explain the difference between low-poly and high-poly models. In which situations would you choose each? (5)
- 3) What is the role of HRTF (Head-Related Transfer Function) in creating spatial audio? (2)
 - A)
 - B) Illustrate the fundamental difference between a 3D Model and a Texture. (3)
 - C) Compare and contrast writing code in C# for Unity versus using Blueprints for Unreal Engine. Outline one advantage and one disadvantage of each visual scripting approach for designers and programmers. (5)
- 4) Identify the primary cause of Simulator Sickness in VR users? (2)
 - A)

- B) How does a "Lighthouse" tracking system differ from a traditional infrared tracking system that uses cameras? (3)
- C) Explain the problem of "drift" in IMU-based tracking and describe how a typical VR system compensates for it using a technique like sensor fusion. (5)
- 5) What does the term "Six Degrees of Freedom (6DoF)" mean? (2)
- A)
- B) Explain the role of Three-Dimensional (3D) Sound in enhancing presence in a virtual environment. (3)
- C) Explain the term "Perception Engineering" and illustrate how it integrates engineering, human physiology, and perception in the design of VR systems (5)

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Event Name		Subject	
058-END SEMSTER-ODD-2025		NPT_4135:Extended Reality Technologies	
#	Course Outcome	Description	Marks
1	2	CO 2	15
2	3	CO 3	15
3	4	CO 4	20
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		NPT_4135:Extended Reality Technologies	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	7(14%)
3	Application	8	31(62%)
4	Analysis	4	12(24%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)